Experiment:-8

1. Aim

To study the operation of differentiator and integrator circuits using an operational amplifier and observe their input and output waveforms through LTspice simulation.

2. Apparatus Required

- Computer or laptop with LTspice installed.
- Operational amplifier model available in LTspice.
- Resistors and capacitors of suitable values.
- Function generator or voltage source for input signals.
- DC power supplies for the operational amplifier.

3. Theory

An operational amplifier can be configured to perform mathematical operations on an input signal. Two important applications of an op-amp are differentiation and integration.

3.1. Differentiator Amplifier

A differentiator is an operational amplifier circuit in which the output voltage is proportional to the rate of change of the input voltage with respect to time. In an ideal differentiator, a capacitor is connected at the input and a resistor is connected in the feedback path. When the input voltage changes, the capacitor current changes, producing an output voltage proportional to the derivative of the input signal. A differentiator produces an output according to the rate at which the input signal changes. For example, a square-wave input produces positive and negative pulses at its transitions.

3.2. Integrator Amplifier

An integrator is an operational amplifier circuit in which the output voltage is proportional to the integral of the input voltage with respect to time. In an ideal integrator, a resistor is connected at the input and a capacitor is connected in the feedback path. The capacitor stores charge according to the input current, resulting in an output voltage proportional to the accumulated input signal. For example, when a square-wave input is applied to an integrator, the output waveform is approximately triangular.

4. Mathematical Equations

4.1. Differentiator Amplifier

The output voltage of an ideal inverting differentiator is:

$$\boxed{V_{out} = -RC\frac{dV_{in}}{dt}}$$

where:

- R is the feedback resistance.
- C is the input capacitance.
- V_{in} is the input voltage.
- V_{out} is the output voltage.

The negative sign indicates the phase inversion of the output.

4.2. Integrator Amplifier

The output voltage of an ideal inverting integrator is:

$$\boxed{V_{out} = -\frac{1}{RC}\int V_{in}\,dt}$$

where:

- R is the input resistance.
- C is the feedback capacitance.
- V_{in} is the input voltage.
- V_{out} is the output voltage.

The negative sign indicates the phase inversion of the output.

5. Differentiator Circuit Diagram

Construct the differentiator amplifier circuit in LTspice using an operational amplifier, input capacitor, and feedback resistor.

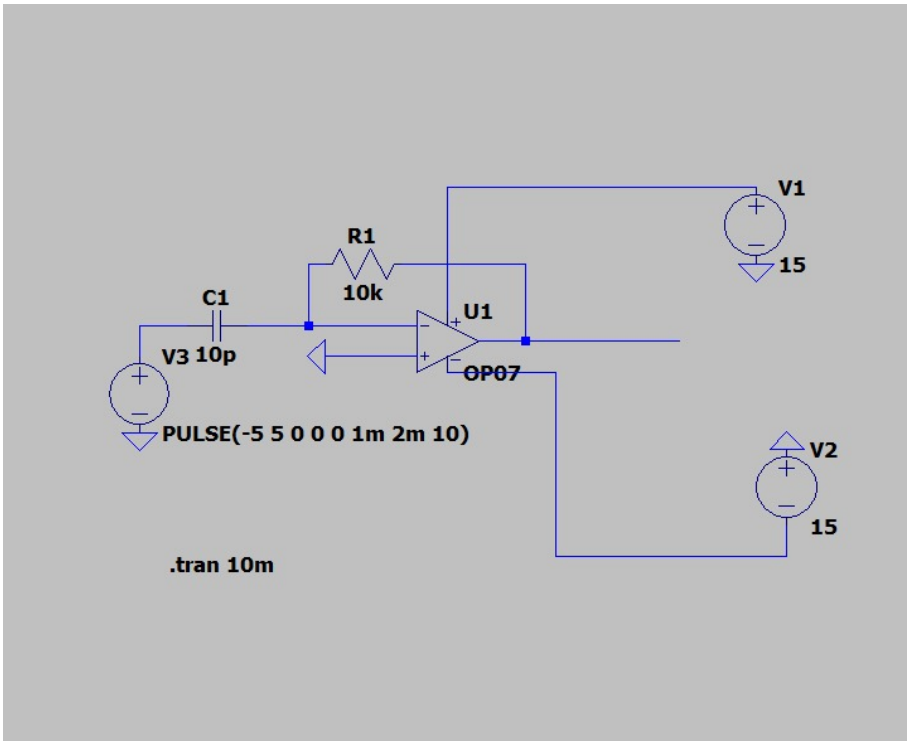


Figure 1: Differentiator amplifier circuit in LTspice

6. Integrator Circuit Diagram

Construct the integrator amplifier circuit in LTspice using an operational amplifier, input resistor, and feedback capacitor.

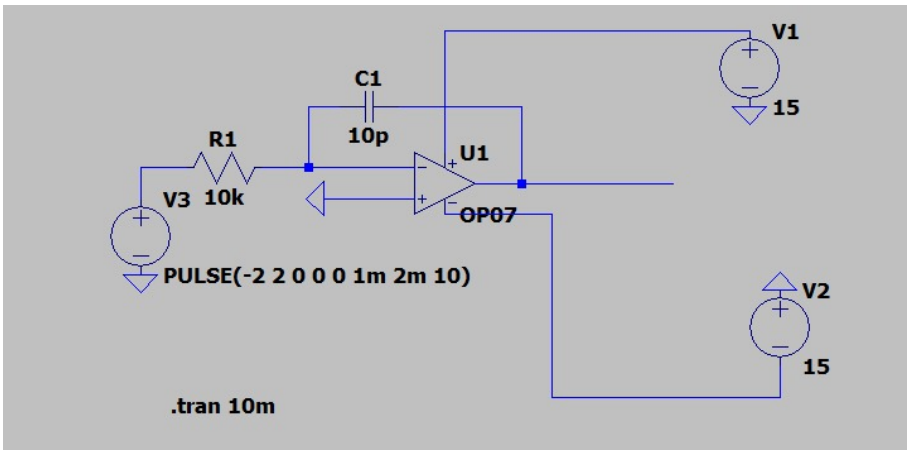


Figure 2: Integrator amplifier circuit in LTspice

7. Procedure

7.1. Differentiator Amplifier

1. Open LTspice and create a new schematic.

2. Select an operational amplifier model from the component library.
3. Connect the operational amplifier in the differentiator configuration using a capacitor at the input and a resistor in the feedback path.
4. Connect the positive and negative DC supply voltages to the operational amplifier.
5. Apply a square-wave input signal using a voltage source.
6. Set suitable input amplitude, frequency, period, and rise and fall times.
7. Configure transient analysis and select an appropriate simulation time.
8. Run the simulation and plot both input and output voltages.
9. Observe the output pulses generated at the rising and falling edges of the input signal.

7.2. Integrator Amplifier

1. Create a new schematic in LTspice.
2. Select an operational amplifier model.
3. Connect the operational amplifier in the integrator configuration using a resistor at the input and a capacitor in the feedback path.
4. Connect the positive and negative DC supply voltages.
5. Apply a square-wave input signal using a voltage source.
6. Set suitable input amplitude, frequency, and simulation time.
7. Configure transient analysis and run the simulation.
8. Plot both input and output voltages.
9. Observe the triangular output waveform obtained by integrating the square-wave input.

8. Differentiator Simulated Waveform

Observe the input and output waveforms of the differentiator amplifier in LTspice. The output should show pulses corresponding to the transitions of the square-wave input.

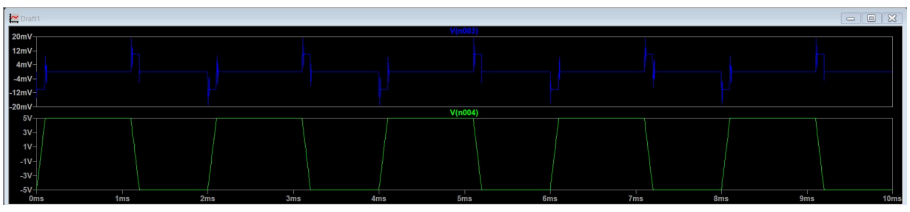


Figure 3: Differentiator input and output waveforms

9. Integrator Simulated Waveform

Observe the input and output waveforms of the integrator amplifier in LTspice. The output should show an approximately triangular waveform when a square-wave input is applied.

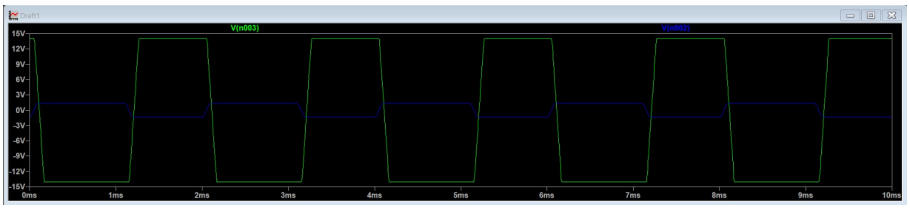


Figure 4: Integrator input and output waveforms

10. Conclusion

The operation of differentiator and integrator amplifiers using an operational amplifier was studied through LTspice simulation. The differentiator produced an output proportional to the rate of change of the input signal, while the integrator produced an output proportional to the integral of the input signal.